Fast Fourier Transform (FFT)

Butterfly Structure

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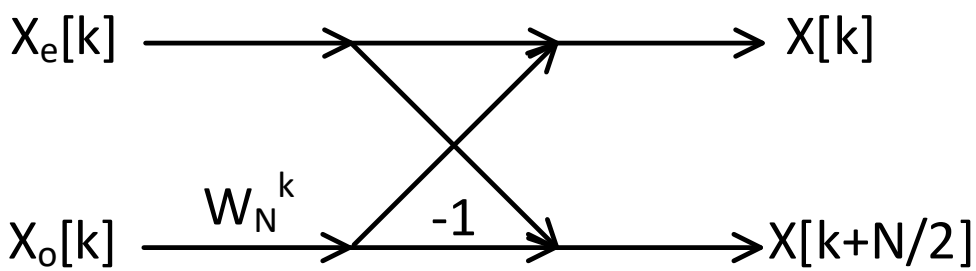
Butterfly Structure

Recall FFT core equations:

- $X[k] = X_e[k] + W_N^k\, X_o[k], k = 0, 1, \ldots, \frac{N}{2} - 1$
- $X\left[k + \frac{N}{2}\right] = X_e[k] - W_N^k\, X_o[k], k = 0, 1, \ldots, \frac{N}{2} - 1$

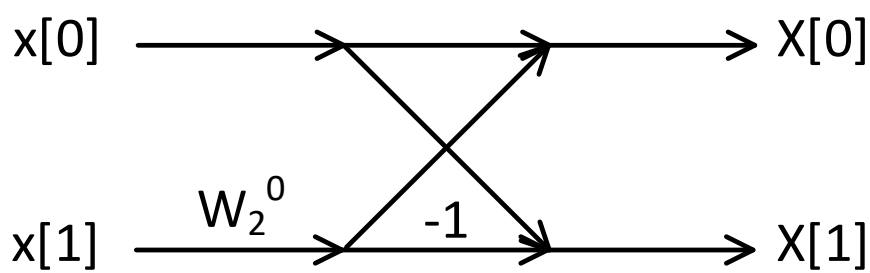
Represent the core equations using a butterfly structure:



- Each butterfly has 1 CMULT and 2 CADDS.
- Each butterfly computes two samples of N-point DFT.
- So N/2 butterflies are needed to compute N-point DFT at any stage.

Butterfly Structure (N = 2)

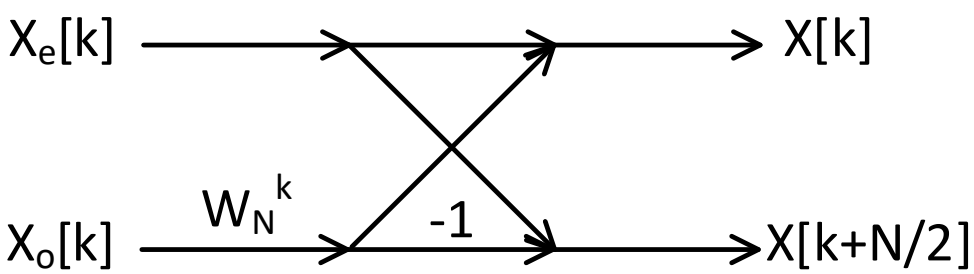
- $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn} = \sum_{n=0}^{1} x[n] e^{-j\frac{2\pi}{2}kn}, k = 0, 1$
- $X[0] = x[0] + x[1] e^{-j\frac{2\pi}{2}0} = x[0] + x[1]$
- $X[1] = x[0] + x[1] e^{-j\frac{2\pi}{2}1} = x[0] - x[1] e^{-j\frac{2\pi}{2}0} = x[0] - x[1]$

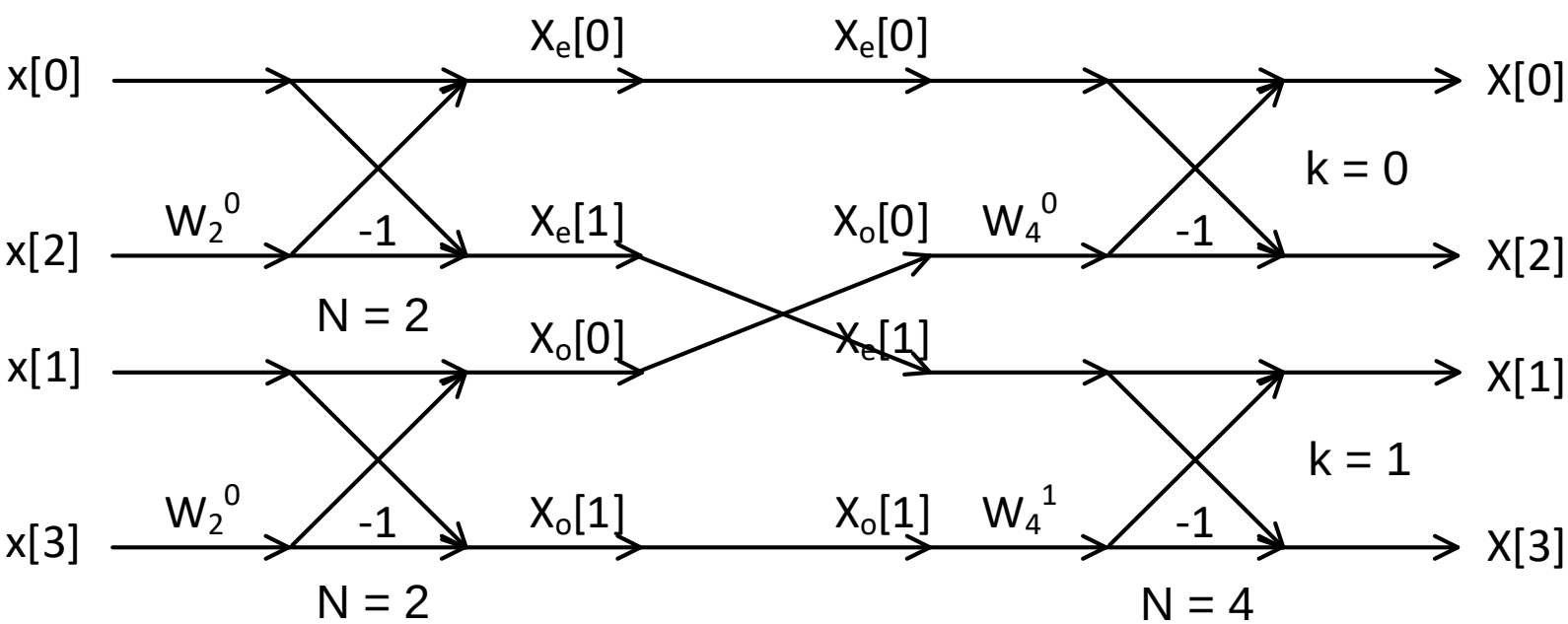


$W_N^k = e^{-j\frac{2\pi}{N}k}$

Butterfly Structure (N = 4)

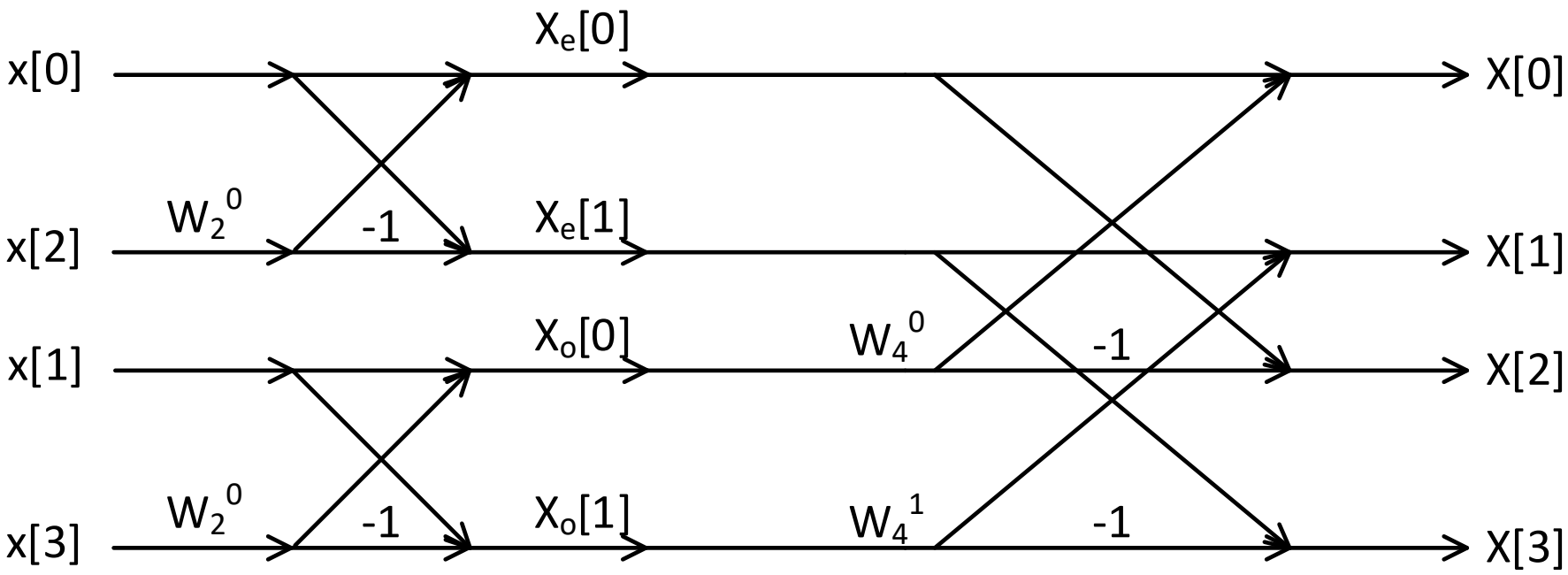
We need $\log_2 N = 2$ stages and $N/2 = 2$ butterflies per stage:





Butterfly Structure (N = 4)

Rearrange the order of the last stage:



Bit Reverse Addressing

- **Bit reverse addressing is used to reorder the input samples x[n]:**

Input Array $x_i[n]$	Input Array Index n	Bit Reverse Index n	Input Array Stores Input Sample
$x_i[0]$	00	00	x[0]
$x_i[1]$	01	10	x[2]
$x_i[2]$	10	01	x[1]
$x_i[3]$	11	11	x[3]

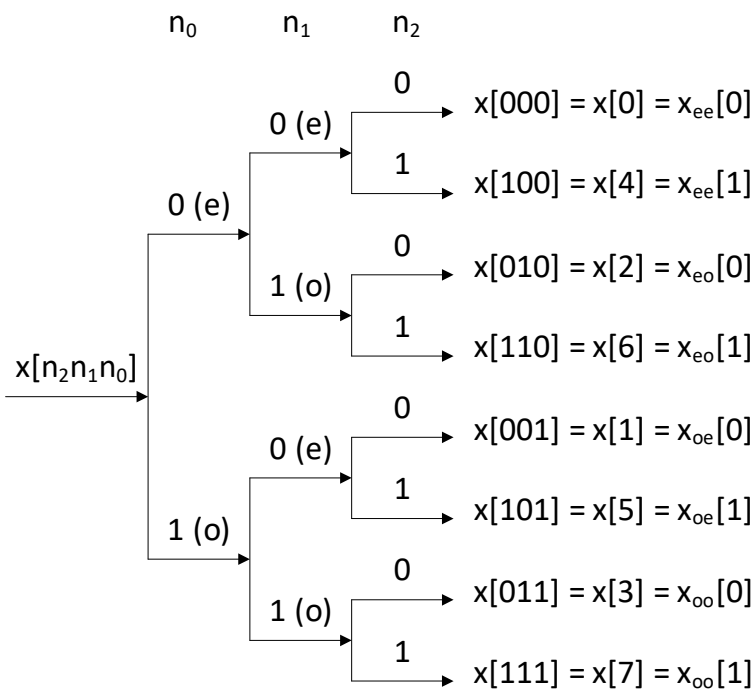
Exercise

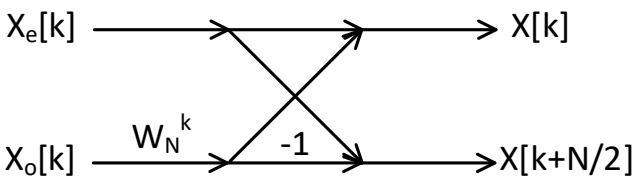
- Put x[0], x[1], …., x[7] in bit reverse order.

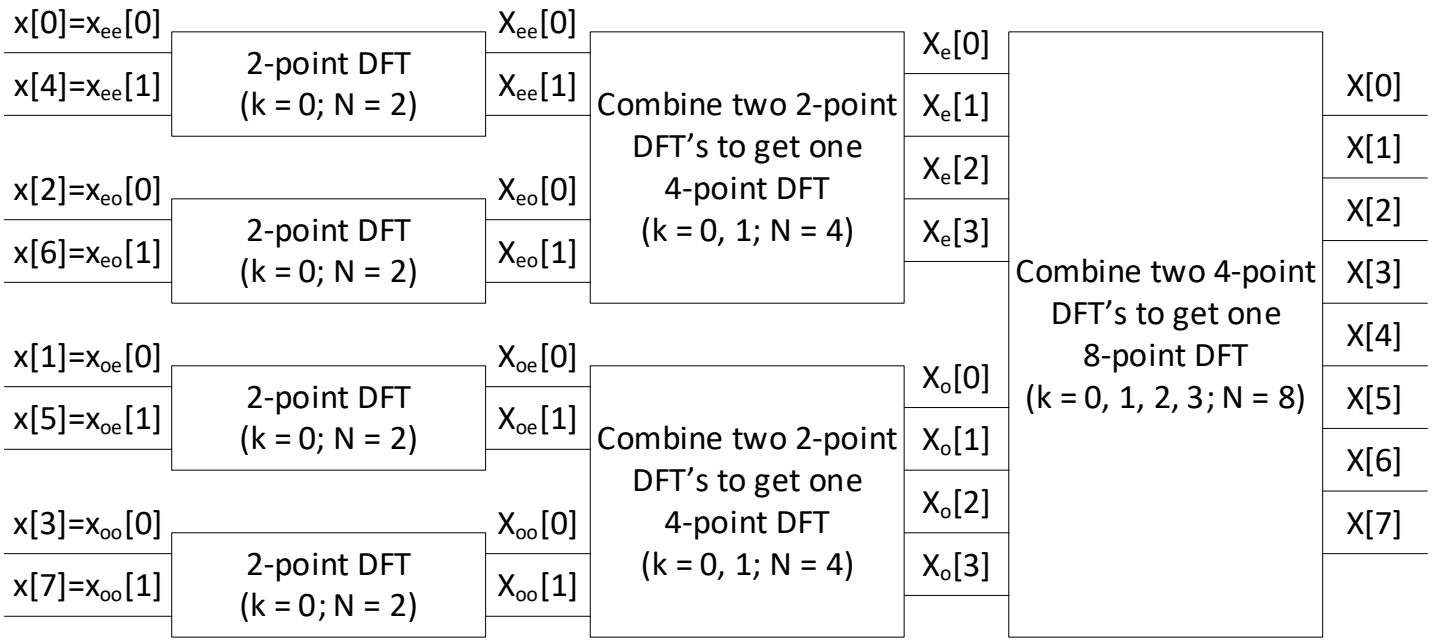
Answer

Input Array $x_i[n]$	Input Array Index n	Bit Reverse Index n	Input Array Stores Input Sample
$x_i[0]$	000	000	x[0]
$x_i[1]$	001	100	x[4]
$x_i[2]$	010	010	x[2]
$x_i[3]$	011	110	x[6]
$x_i[4]$	100	001	x[1]
$x_i[5]$	101	101	x[5]
$x_i[6]$	110	011	x[3]
$x_i[7]$	111	111	x[7]

Butterfly Structure (N = 8)

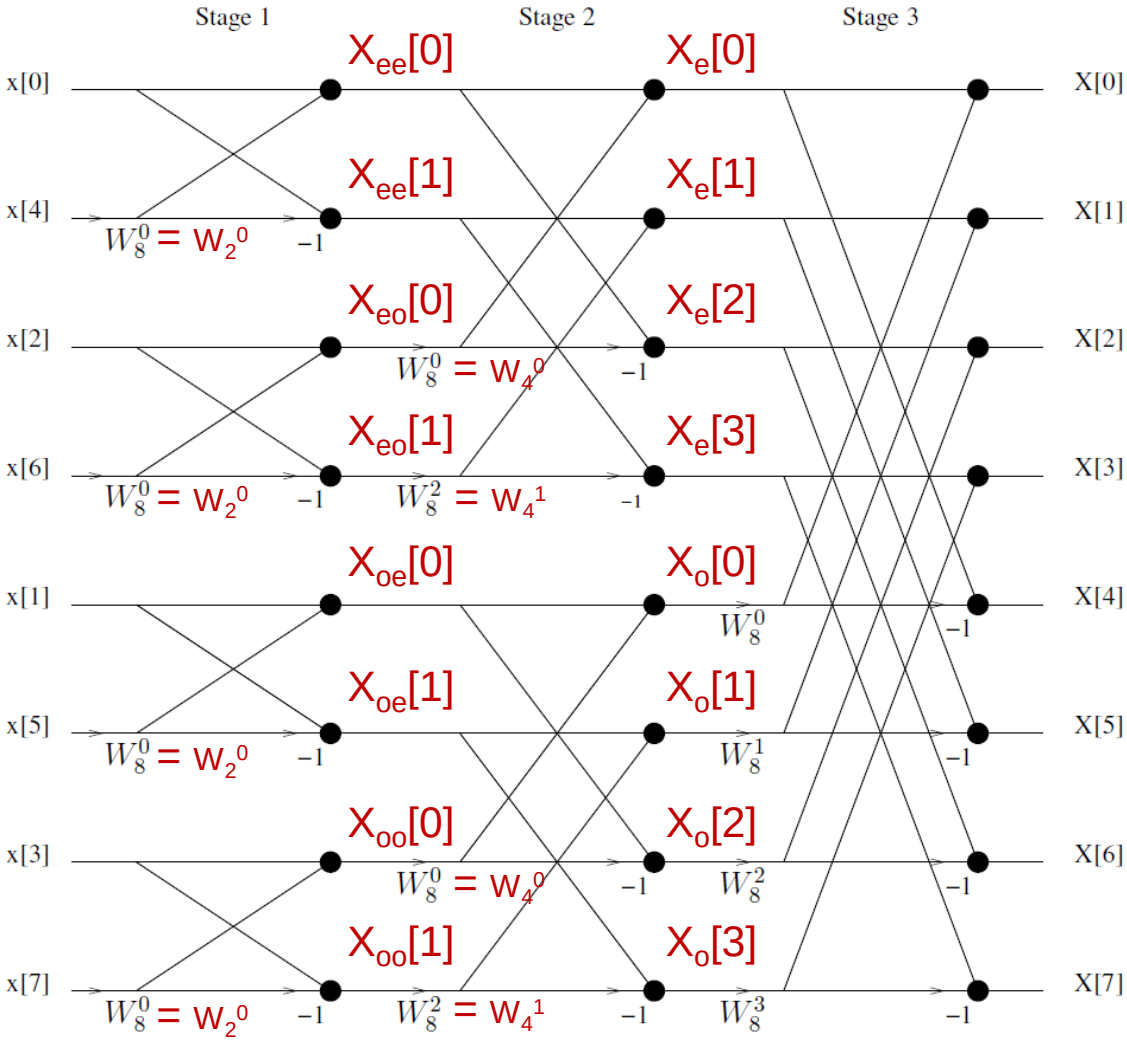






Butterfly Structure (N = 8)

- We need $\log_2 N = 3$ stages and $N/2 = 4$ butterflies per stage:



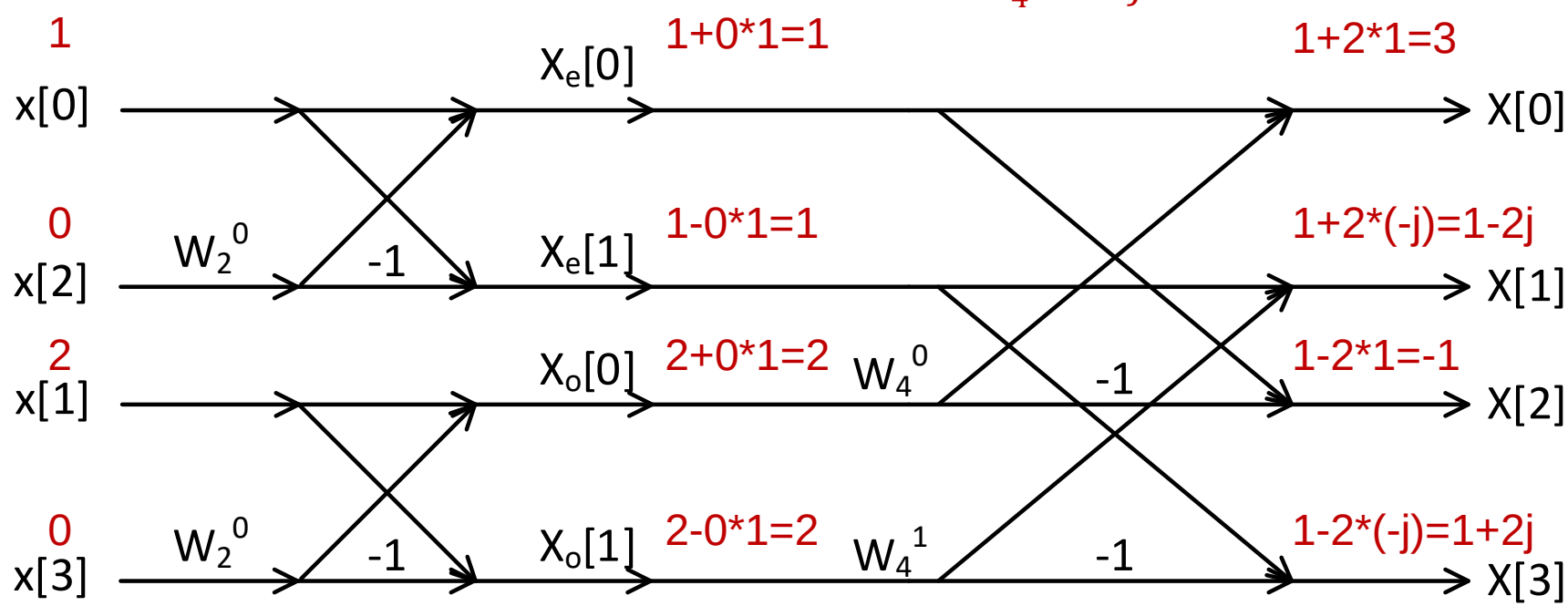
Example

- **Sketch the flow graph of the 4-point FFT computation of x[n] = [1, 2, 0, 0].**

Answer

$$W_N^k = e^{-j\frac{2\pi}{N}k}$$
$$W_2^0 = 1$$
$$W_4^0 = 1$$
$$W_4^1 = -j$$

x[n] = [1, 2, 0, 0]



X[k] = [3, 1-2j, -1, 1+2j]

FFT: Computation Complexity

At each stage, N/2 butterflies are needed.

- Each butterfly has 1 CMULT and 2 CADDs.
- Each stage has N/2 CMULTs and N CADDs.

How many stages are needed?

- For $N = 2^{\alpha}$, $\alpha = \log_2 N$ stages.

Total computations for N-point FFT:

- $(N*\log_2 N)/2$ CMULTs
- $N*\log_2 N$ CADDs

Computation Complexity Comparison: DFT vs FFT

	Direct computation using DFT		Computation using radix 2 FFT	
Number of points	*Complex multiplications*	*Complex additions*	*Complex multiplications*	*Complex additions*
N	N^2	$N^2 - \mathrm{N}$	$\frac{N}{2} log_2(N)$	$N log_2(N)$
4	16	12	4	8
16	256	240	32	64
64	4096	4032	192	348
256	65536	65280	1024	2048
1024	1048576	1047552	5120	10240

Number of Computations for FFT

Total computations for N-point FFT (simplified):

- $(N*\log_2 N)/2$ CMULTs, $N*\log_2 N$ CADDs
- 1 CMULT = 4 MULTs + 2 ADDs
- Exclude CADDs and ADDs, less computation intensive compared to multiplications
- $2N*\log_2 N$ MULTs

Example Revisited

- **Consider 5 minutes of speech sampled at 8KHz:**
- **Length of discrete domain samples x[n] is**
  - $N = 5 * 60 * 8000 = 2.4 * 10^6$ samples
- **Number of computations (simplified):**
  - N-point DFT: N^2 CMULTs = $4N^2$ MULTs = $4 * 2.4^2 * 10^{12}$ MULTs
  - FFT: $(N * \log_2 N) / 2$ CMULTs = $2N * \log_2 N$ MULTs = $1.017 * 10^8$ MULTs
- **DSP at 168MHz with 1 instruction per clock cycle, i.e., 168MIPS**
- **Time to compute N-point DFT for 5 minutes of speech:**
  - DFT: $4 * 2.4^2 * 10^{12} / (168 * 10^6) \approx 137143$ sec ≈ 38 hr
  - FFT: $1.017 * 10^8 / (168 * 10^6) \approx 0.6$ sec

Question

For a N = 64-point DFT, how many real multiplication operations are needed if it is implemented using FFT?

a) 448

b) 768

c) 4096

d) 16384

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